Example 2.38

By the principle of mathematical induction prove that $n^2 + n$ is an even number, for all $n \in N$.

Solution

Let $P(n)$ denote the statement that, "$n^2 + n$ is an even number".

Put $n = 1$

$\therefore 1^2 + 1 = 1 + 1 = 2$, an even number.

Let us assume that $P(k)$ is true.

$\therefore k^2 + k$ is an even number is true

$\therefore$ Take $k^2 + k = 2m$ ...(1)

To prove $P(k + 1)$ is true

$\therefore (k+1)^2 + (k+1)$

$= k^2 + 2k + 1 + k + 1$

$= k^2 + k + 2k + 2$

$= 2m + 2(k + 1)$ by (1)

$= 2(m + k + 1)$ (a multiple of 2)

$\therefore (k + 1)^2 + (k + 1)$ is an even number

$\therefore P(k + 1)$ is true whenever $P(k)$ is true.

$P(n)$ is true for $n \in N$.

Exercise 2.5

By the principle of mathematical induction, prove the following

1. $1^3 + 2^3 + 3^3 + \ldots + n^3 = \frac{n^2(n+1)^2}{4}$ for all $n \in N$.
2. $1.2 + 2.3 + 3.4 + \ldots + n(n + 1) = \frac{n(n+1)(n+2)}{3}$, for all $n \in N$.
3. $4 + 8 + 12 + \ldots + 4n = 2n(n+1)$, for all $n \in N$.
4. $1 + 4 + 7 + \ldots + (3n - 2) = \frac{n(3n-1)}{2}$, for all $n \in N$.
5. $3^{2n} - 1$ is a divisible by 8, for all $n \in N$.
6. $a^n - b^n$ is divisible by $a - b$, for all $n \in N$.
7. $5^{2n} - 1$ is divisible by 24, for all $n \in N$.
8. $n(n + 1)(n+2)$ is divisible by 6, for all $n \in N$.
9. $2^n > n$, for all $n \in N$.

2.5 Binomial Theorem

An algebraic expression of sum or the difference of two terms is called a binomial. For example $(x+y), (5a-2b), \left(x+\frac{1}{y}\right), \left(p+\frac{5}{p}\right), \left(\frac{7}{4}+\frac{1}{y^2}\right)$ etc., are binomials. The Binomial theorem or Binomial Expression is a result of expanding the powers of binomials. It is possible to expand $(x + y)^n$ into a sum involving terms of the form ax^by^c, exponents b and c are non-negative integers with $b + c = n$, the coefficient 'a' of each term is a positive integer called binomial coefficient.

Expansion of $(x + a)^2$ was given by Greek mathematician Euclid on 4th century and there is an evidence that the binomial theorem for cubes i.e., expansion of $(x+a)^3$ was known by 6th century in India. The term Binomial coefficient was first introduced by Michael Stifle in 1544. Blaise Pascal (19 June 1623 to 19 August 1662) a French Mathematician, Physicist, inventor, writer and catholic theologian. In his Treatise on Arithmetical triangle

02_11th_BM-STAT_Ch-2-EM.indd 41 21-04-2020 12:18:40 PM